

7 (a) (i)	six vertical lines from plate to plate equally spaced across plates [only allow if greatest to least spacing is < 1.3 , condone slight curving on the two edges. There must be no area between the plates where an additional line(s) could be added.] arrow downwards on at least one line	B1	
(ii)	$E = V / d$ $= 1200 / 40 \times 10^{-3} = 3.0 \times 10^4 \text{ V m}^{-1}$ (allow 1 s.f.)	C1	
		A1	[2]
(b) (i)	$F = Ee$ $= 3 \times 10^4 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-15} \text{ N}$	C1	
(ii)	couple = $F \times$ separation of charges $= 4.8 \times 10^{-15} \times 15 \times 10^{-3} = 7.2 \times 10^{-17}$ unit: N m or unit consistent with unit used for the separation	C1	
		A1	
		B1	[3]
(iii)	A at top/next to +ve plate B at bottom/next to -ve plate vertically aligned [could be shown on the diagram] forces are equal and opposite in same line / no resultant force and no resultant torque	M1	
		A1	[2]